

SERVER SCRIPT

SHIFT PRODUCTION SUBMIT AUTO-DEDUCT

Script Type:	DocType Event
Reference Document Type:	Shift Wise Production Entry
Event Frequency:	All
DocType Event:	Before Submit

Script

```
# Server Script for Shift Wise Production Entry Submission
# Imports are not allowed in Frappe Server Scripts. 'frappe' is global.

try:
    def get_or_create_patty_item(quality, colour, gsm, width_inch):
        # 1. Mapping Tables
        QUALITY_MAP = {
            "PREMIUM": "100", "PLATINUM": "101", "SUPER PLATINUM": "102", "GOLD": "103",
            "SILVER": "104", "BRONZE": "105", "CLASSIC": "106", "SUPER CLASSIC": "107",
            "LIFE STYLE": "108", "ECO SPECIAL": "109", "ECO GREEN": "110", "SUPER ECO": "111",
            "ULTRA": "112", "DELUXE": "113", "UV": "114"
        }
        COLOUR_MAP = {
            "BRIGHT WHITE": "001", "MILKY WHITE": "002", "SUPER WHITE": "003", "SUNSHINE WHITE": "004",
            "BLEACH WHITE 1.0": "005", "BLEACH WHITE 2.0": "006", "BRIGHT IVORY": "101", "IVORY 2.0": "102",
            "CREAM 1.0": "121", "LEMON YELLOW": "142", "LEMON YELLOW 2.0": "143", "LEMON YELLOW 4.0": "1
45",
            "GOLDEN YELLOW": "161", "GOLDEN YELLOW 5.0": "166", "BRIGHT ORANGE": "181", "DARK ORANG
E": "182",
            "ORANGE 1.0": "183", "BABY PINK": "201", "DARK PINK": "204", "PINK 4.0": "206",
            "CRIMSON RED": "221", "RED 1.0": "222", "LIGHT MAROON": "241", "RED 2.0": "242",
            "DARK MAROON": "251", "MEDICAL BLUE": "261", "BLUE 3.0": "283", "BLUE 5.0": "301",
            "PEACOCK BLUE": "302", "ROYAL BLUE": "322", "NAVY BLUE": "344", "NAVY BLUE 2.0": "347",
            "VIOLET": "361", "PARROT GREEN": "401", "RELIANCE GREEN": "421", "SEA GREEN": "446",
            "LIGHT GREY": "464", "DARK GREY": "465", "CHOCOLATE BROWN": "487", "LIGHT BEIGE": "501",
            "DARK BEIGE": "521", "BLACK": "542", "WHITE MIX": "600", "BLACK MIX": "601",
            "COLOR MIX": "602", "BEIGE MIX": "603"
        }

        # 2. Extract and Normalize Data
        qual_name = str(quality or "").strip().upper()
        col_name = str(colour or "").strip().upper()

        # GSM Parsing
        gs_raw = str(gsm or "0")
        gs_digits = "".join([c for c in gs_raw if c.isdigit() or c == "."])
        gs_val = float(gs_digits) if gs_digits else 0.0
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gs_int = int(gs_val)

# Width Processing (Inch to MM with 5mm rounding)
w_inch = float(width_inch or 0)
# 1 inch = 25.4 mm. User wants rounding to nearest 5.
w_mm_raw = w_inch * 25.4
w_mm = int(round(w_mm_raw / 5.0) * 5)

# 3. Generate Codes
process_code = "100"
quality_code = QUALITY_MAP.get(qual_name, "000")
colour_code = COLOUR_MAP.get(col_name, "000")
gsm_code = str(gs_int).zfill(3)
mm_code = str(w_mm).zfill(4)

ic = process_code + quality_code + colour_code + gsm_code + mm_code

# 4. Generate Name
# patty : NON WOVEN FABRIC [QUALITY] [COLOUR] [GSM] GSM W - [WIDTH_INCH]" ( [WIDTH_MM] MM
)
# Note: Use whatever gsm string was passed for the name as per user request
gs_name_str = str(gs_int) # Defaulting to int for cleaner name if possible
if gs_val % 1 != 0: gs_name_str = str(gs_val) # keep decimal if it exists

inm = "patty : NON WOVEN FABRIC " + qual_name + " " + col_name + " " + gs_name_str + " GSM W - " + str
(w_inch) + "\" ( " + str(w_mm) + " MM )"

# 5. HSN Logic
hsn = "56031400" # Default > 150
if gs_int <= 24: hsn = "56031100"
elif 25 <= gs_int <= 70: hsn = "56031200"
elif 71 <= gs_int <= 150: hsn = "56031300"

# 6. Create Item if missing
if not frappe.db.exists("Item", ic):
    frappe.get_doc({
        "doctype": "Item",
        "item_code": ic,
        "item_name": inm,
        "item_group": "Products",
        "stock_uom": "Kg",
        "is_stock_item": 1,
        "valuation_rate": 0.0,
        "standard_rate": 0.0,
        "gst_hsn_code": hsn,
        "default_warehouse": "Raw Materials - JSB-1ZT",
        "allow_zero_valuation_rate": 1
```

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allow_zero_validation_rate : 1
    }).insert(ignore_permissions=True)
    return [ic, inm]

# Global Context Setup
comp = doc.get("company")
if not comp:
    comp_list = frappe.get_all("Company", limit=1)
    if comp_list: comp = comp_list[0].name

# 1. Get Batch No
rf = {"docstatus": 1, "posting_date": doc.posting_date, "shift": doc.shift}
if doc.get("unit"): rf["unit"] = doc.unit
bn = doc.get("batch_no") or ""
rpn = frappe.db.get_value("Roll Production Entry", rf, "name")
if rpn:
    fb = frappe.db.get_value("Roll Production Entry", rpn, "shift_batch_number")
    if fb: bn = str(fb)

# 2. Process Wastage (Added to Patty Stock)
w_table_name = "running_patty_wastage" if doc.get("running_patty_wastage") else "running_patty_waste"
w_rows = doc.get(w_table_name) or []

waste_receipt_items = []
for w in w_rows:
    # Only process if both width and qty are provided
    wq = float(w.get("wastage_qty_kg") or 0)
    win = float(w.get("width_inch") or 0)

    if wq > 0 and win > 0:
        qual = w.get("quality")
        col = w.get("colour")
        gs = w.get("gsm")

        p_res = get_or_create_patty_item(qual, col, gs, win)
        p_ic = p_res[0]
        p_inm = p_res[1]

        ps = frappe.new_doc("Patty Stock")
        ps.item_code = p_ic
        ps.item_name = p_inm
        ps.quality = qual
        ps.colour = col
        ps.gsm = gs
        ps.width_inch = win
        ps.balance_quantity = wq
        ps.uom = "Kg"
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ps.company = comp
ps.insert(ignore_permissions=True)

waste_receipt_items.append({"item_code": p_ic, "qty": wq})

# Automate Material Receipt for Wastage
if waste_receipt_items:
    sw = "Raw Materials - JSB-1ZT"
    if not frappe.db.exists("Warehouse", sw):
        abbr = str(frappe.db.get_value("Company", comp, "abbr")) if comp else ""
        test_sw = "Stores - " + abbr
        sw = test_sw if frappe.db.exists("Warehouse", test_sw) else "Raw Materials - " + abbr

    sr = frappe.new_doc("Stock Entry")
    sr.purpose = "Material Receipt"
    sr.stock_entry_type = "Material Receipt"
    sr.company = comp
    for it in waste_receipt_items:
        row = sr.append("items", {})
        row.item_code = it["item_code"]
        row.qty = it["qty"]
        row.t_warehouse = sw
        row.allow_zero_valuation_rate = 1
        row.valuation_rate = 0.0
    sr.insert(ignore_permissions=True)
    sr.submit()

# 3. Deduct Consumption (Recycle)
ri = []
ct = doc.get("recycle") or doc.get("patty_consumption") or []
for r in ct:
    std_q = float(r.get("standard_consumption") or 0)
    cq = float(r.get("consumption_qty_kg") or r.get("actual_consumption") or r.get("available_balance") or 0)
    if cq > 0:
        qual = str(r.get("quality") or "")
        col = str(r.get("colour") or "")
        gs = str(r.get("gsm") or "")
        win = str(r.get("width_inch") or "")

        p_res = get_or_create_patty_item(qual, col, gs, win)
        p_ic = p_res[0]

    as_list = frappe.get_all("Patty Stock",
        filters={"quality": qual, "colour": col, "gsm": gs, "width_inch": win, "balance_quantity": [ ">", 0 ]},
        fields=["name", "balance_quantity"],
        order_by="creation asc")
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)

rem = cq
for s in as_list:
    if rem <= 0: break
    d_amt = min(s.balance_quantity, rem)
    new_bal = s.balance_quantity - d_amt

    if new_bal <= 0:
        frappe.delete_doc("Patty Stock", s.name, ignore_permissions=True)
    else:
        frappe.db.set_value("Patty Stock", s.name, "balance_quantity", new_bal)
    rem = rem - d_amt

if rem > 0:
    frappe.throw("Insufficient stock for " + qual + " " + col + ". Need " + str(cq) + " but shortfall " + str(rem))
ri.append({
    "item_code": p_ic,
    "std_qty": min(cq, std_q),
    "extra_qty": max(0, cq - std_q)
})
```

4. Process Item Consumption Variance

For regular items, we check actual vs standard.

If actual > standard: Material Issue of the difference.

If actual < standard: Material Receipt of the difference.

cv_issue = []

cv_receipt = []

ci_rows = doc.get("consumption_items") or []

for ci in ci_rows:

std_q = float(ci.get("standard_consumption") or 0)

act_q = float(ci.get("actual_consumption") or 0)

diff = act_q - std_q

ic_q = ci.get("item_code")

if diff > 0: # Extra consumption

cv_issue.append({"item_code": ic_q, "qty": diff})

elif diff < 0: # Less consumption

cv_receipt.append({"item_code": ic_q, "qty": abs(diff)})

Final Stock Entries

sw = "Raw Materials - JSB-1ZT"

if not frappe.db.exists("Warehouse", sw):

abbr = str(frappe.db.get_value("Company", comp, "abbr")) if comp else ""

test_sw = "Stores - " + abbr

sw = test_sw if frappe.db.exists("Warehouse", test_sw) else "Raw Materials - " + abbr

```
# 1. Consolidated Material Issue
# Includes: Patty Consumption (Standard + Extra) AND Regular Item Extra Consumption
si = frappe.new_doc("Stock Entry")
si.purpose = "Material Issue"
si.stock_entry_type = "Material Issue"
si.company = comp
si.remarks = "Shift Consumption Adjustments"

# Add Patty pieces (from ri)
for it in ri:
    qty_to_issue = float(it.get("std_qty") or 0) + float(it.get("extra_qty") or 0)
    if qty_to_issue > 0:
        row = si.append("items", {})
        row.item_code = it["item_code"]
        row.qty = qty_to_issue
        row.s_warehouse = sw
        row.allow_zero_valuation_rate = 1
        row.valuation_rate = 0.0

# Add Regular Item Extra Consumption (from cv_issue)
for it in cv_issue:
    row = si.append("items", {})
    row.item_code = it["item_code"]
    row.qty = it["qty"]
    row.s_warehouse = sw
    row.allow_zero_valuation_rate = 1
    row.valuation_rate = 0.0

if si.items:
    si.insert(ignore_permissions=True)
    si.submit()

# 2. Consolidated Material Receipt for Reduced Consumption
if cv_receipt:
    sr_var = frappe.new_doc("Stock Entry")
    sr_var.purpose = "Material Receipt"
    sr_var.stock_entry_type = "Material Receipt"
    sr_var.company = comp
    sr_var.remarks = "Reduced Consumption Adjustment"
    for it in cv_receipt:
        row = sr_var.append("items", {})
        row.item_code = it["item_code"]
        row.qty = it["qty"]
        row.t_warehouse = sw
        row.allow_zero_valuation_rate = 1
```

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row.valuation_rate = 0.0
sr_var.insert(ignore_permissions=True)
sr_var.submit()
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except Exception as e:
    frappe.log_error(title="shift_production_submit error", message=str(e))
    frappe.throw("Inventory Update Error: " + str(e))
```

Request Limit: 5

Time Window (Seconds): 86400